

Answer Key — Biology Grade 11

Sub-Strand 1.2: Ecology • Lesson 9: Pollution, Garbage Rivers and Restoring Our River: The Final Explanation

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
A	D	D	B	B	B	D	D	D	A

USE AFTER: PREDICT PHASE — Predict: arrow sketch from the shopping centre to the river

Q1. Before a walk to the shopping centre, learners predict where the water from drains goes when it rains. Which statement best describes the likely pathway of pollutants such as oil, soapy water and litter from a shopping centre?

- A. They are washed by rain into drains and carried to a nearby river**
- B. They evaporate completely and leave no pollution behind
- C. They are absorbed by the drains and disappear
- D. They stay on the ground at the shopping centre forever

Correct answer: A. Rain runoff and blocked or open drains carry oil, detergent and litter from towns into rivers such as the Nairobi River, which is why the walk traced pollutant destinations.

USE AFTER: OBSERVE PHASE — Observe: walk to the neighbouring shopping centre and observation table

Q2. During a pollution walk at a Kenyan shopping centre, learners see these items: plastic bottles, smoke from burning rubbish, hooting matatus, and soapy water from a car wash flowing into a drain. Which item is correctly matched to its type of pollution?

- A. Soapy water in a drain: solid waste pollution
- B. Plastic bottles: air pollution
- C. Smoke from burning rubbish: water pollution
- D. Hooting matatus: noise pollution**

Correct answer: D. Hooting is unwanted loud sound, so it is noise pollution. Plastic bottles are solid waste, smoke is air pollution and soapy water is water pollution.

USE AFTER: OBSERVE PHASE — Observe: teacher notes misconceptions such as 'only factories pollute'

Q3. A learner says, 'Only factories pollute rivers, so ordinary people in town do not cause river pollution.' Which observation from a shopping centre walk best shows that this idea is wrong?

- A. Factories release smoke into the air
- B. Fish need oxygen to breathe
- C. Some rivers are very long
- D. Soapy water from car washes and litter flow into drains that lead to the river**

Correct answer: D. Car wash detergent and litter from everyday activities enter drains and reach rivers, so ordinary people and small businesses also pollute.

USE AFTER: EXPLAIN PHASE — Explain: Claim-Evidence-Reasoning Final Explanation (nutrient input)

Q4. In the Nairobi River, effluent and fertiliser runoff add nitrates and phosphates to the water. What is the first effect of these added nutrients?

- A. Fish grow faster because the water is clean
- B. Algae and aquatic weeds grow much faster**
- C. Bacteria stop decomposing dead matter
- D. The water gets more dissolved oxygen immediately

Correct answer: B. Nitrates and phosphates are nutrients that fertilise algae and weeds, causing an algal bloom (eutrophication). Low oxygen comes later.

USE AFTER: EXPLAIN PHASE — Explain: Reasoning part of the CER frame (bacterial decomposition and low oxygen)

Q5. After an algal bloom, the algae die and sink. Why does the amount of dissolved oxygen in the river fall?

- A. Fish breathe out oxygen into the water
- B. Bacteria decomposing the dead algae use up the oxygen**
- C. Nitrates change oxygen into carbon dioxide gas
- D. Dead algae release large amounts of oxygen into the water

Correct answer: B. Decomposer bacteria respire aerobically while breaking down the dead algae and use up dissolved oxygen, leaving too little for fish.

USE AFTER: EXPLAIN PHASE — Explain: 'Which piece of evidence best supports your claim about the smell?'

Q6. The river water sometimes has an unpleasant smell. Which explanation is best supported by the Final Explanation?

- A. Nitrates and phosphates are gases with a bad smell
- B. Decomposition without enough oxygen produces foul-smelling gases**
- C. Cold water always smells bad
- D. Fish release a strong smell when they are happy

Correct answer: B. When oxygen is low, anaerobic decomposition of dead algae, weeds and fish produces foul-smelling gases.

USE AFTER: EXPLAIN PHASE — Explain: linking burning waste and fossil fuels to greenhouse gases and climate change

Q7. Burning rubbish and fossil fuels in towns is linked to global warming. Which statement explains the link, and how it can affect fish in a river?

- A. Burning has no effect on the climate or on rivers
- B. Burning makes more oxygen dissolve in river water, so fish suffocate
- C. Burning removes carbon dioxide from the air, which cools rivers and kills fish
- D. Burning releases greenhouse gases that warm the Earth, and warmer water holds less dissolved oxygen**

Correct answer: D. Burning releases greenhouse gases such as carbon dioxide, which trap heat. Warmer water holds less dissolved oxygen, which stresses fish.

USE AFTER: MODEL BUILDING PHASE — Model: community solutions and poster to restore the river

Q8. A community wants to restore a polluted river where fish have disappeared. Which action matches the cause of the algal bloom best?

- A. Add more fish to the river without changing anything else
- B. Plant trees on a hill far from the river
- C. Hoot loudly near the river to chase away the algae
- D. Reduce fertiliser and detergent use and treat effluent before it reaches the river**

Correct answer: D. The bloom is caused by nitrates and phosphates from fertiliser, detergent and effluent, so cutting these inputs and treating effluent tackles the cause.

USE AFTER: END OF LESSON — Model: comparing the Lesson 1 model with the final model

Q9. In Lesson 1, a learner's model said that fish were disappearing because they were old or had no food. After the final model, which change in thinking shows the best use of evidence?

- A. Fish are lost because algae and weeds give them too much food
- B. Fish are lost because the smell attracts predators
- C. Fish are lost only because the river is too long
- D. Fish are lost mainly because nutrient pollution leads to algal bloom, decomposition and low dissolved oxygen**

Correct answer: D. The final model replaces the first ideas with the evidence-based chain: nutrient input, algal bloom, bacterial decomposition, low oxygen, fish loss and smell.

Q10. Exit ticket: Why do fish decline when algae increase in a river?

- A. Dead algae are decomposed by bacteria that use up the dissolved oxygen, so fish suffocate or leave**
- B. Algae eat the fish directly
- C. Algae add extra oxygen that is harmful to fish
- D. Algae make the water too cold for fish

Correct answer: A. A large amount of dead algae feeds decomposing bacteria that use up dissolved oxygen, so fish cannot breathe and die or move away.